

The Complex Transverse Helmholtz Equation

Variable Separation, Eigen-Narratives, and Parameter Interplay

1. 3D Wave Equation & Variable Separation

In a waveguide, propagation along the longitudinal axis (x) and confinement in the transverse plane (y,z) can be mathematically separated. We start from the 3D Helmholtz equation for the electric field phasor $E(x,y,z)$:

$$\left[\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right] E + (k_0^2 * \epsilon_{\text{tilde}}) E = 0$$

Where $k_0 = \omega/c$ is the free-space wavenumber, and $\epsilon_{\text{tilde}}(y,z)$ is the complex relative permittivity. Assuming a longitudinally uniform waveguide structure, ϵ_{tilde} depends only on (y,z) . We can separate the wave variables as:

$$E(x,y,z) = E_{1d}(x) * E_{2d}(y,z) = \exp(i * \beta_{\text{tilde}} * x) * \Psi(y,z)$$

Substituting this separation into the 3D wave equation yields two decoupled equations:

1. Longitudinal propagation: $\frac{d^2 E_{1d}}{dx^2} = -\beta_{\text{tilde}}^2 * E_{1d}$
2. Transverse Helmholtz equation: $\left[\frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right] \Psi + k_0^2 * \epsilon_{\text{tilde}} * \Psi = \beta_{\text{tilde}}^2 * \Psi$

2. The Eigenvalue and Eigenvector Narrative

The transverse Helmholtz equation can be cast directly into a linear eigenvalue problem:

$$\hat{H}(\omega) \Psi = \lambda * \Psi$$

Where:

- $\hat{H}$ (The Operator): $\hat{H} = \left[\frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right] + k_0^2 * \epsilon_{\text{tilde}}(y,z)$. Because the permittivity ϵ_{tilde} is complex-valued, this operator is non-Hermitian.
- Ψ (The Eigenvector): Represents the spatial transverse profile (mode shape) of the waveguide mode. It defines the spatial distribution of electromagnetic energy.
- λ (The Eigenvalue): $\lambda = \beta_{\text{tilde}}^2 = (\beta_r + i * \beta_i)^2$. It is the square of the complex propagation constant. The real part of the eigenvalue governs wave velocity; the imaginary part governs spatial amplification or decay.

3. Parameter Interplay & Feedback Loop

The three core components - angular frequency (ω), propagation constant (β_{tilde}), and complex permittivity (ϵ_{tilde}) - interact through a self-consistent physical loop:

1. Angular Frequency (ω) tunes the Operator: ω scales the wavenumber $k_0 = \omega/c$, setting

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the baseline spatial frequency of the waves. It also aligns with the transition energy of the active region, determining the frequency range of the gain spectrum $g(\omega)$.

2. Complex Permittivity (ϵ_{tilde}) determines the Eigen-Elements:

- Real Permittivity (ϵ_r) acts as a potential well. A high-index core guides the eigenvector, restricting the mode profile $\Psi(y,z)$.
- Imaginary Permittivity (ϵ_i) contains gain/loss profiles. It is localized in the core (negative ϵ_i represents gain) and distributed in the cladding (positive ϵ_i represents loss/absorptions). This spatial mismatch makes the operator non-Hermitian and curves the phase front of the eigenvector.

3. Complex Propagation Constant (β_{tilde}) describes the Mode State:

- β_r (real part) defines the effective index $n_{\text{eff}} = \beta_r / k_0$ and phase velocity $v_p = \omega / \beta_r$. The largest β_r belongs to the fundamental mode, representing the most deeply bound state.
- β_i (imaginary part) is the net spatial gain/loss coefficient. It is determined by the overlap of the eigenvector Ψ with the imaginary permittivity profile ϵ_i .

Status: Verified Theoretical Framework

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